

Every Natural Strengthening Examined Is False: Type Mismatch in Lossless Reformulations of the Lonely Runner Conjecture

A sharp negative result with a mechanism: four settings across two reformulation worlds—the finite covering reformulation, the transfer-matrix cascade, character rigidity, and the Lonely Runner zonotope—in which every strengthening that would open a new toolkit and was examined is false, with exact rational witnesses ($4183/12288 > 1/3$ at the harmonic instance, from $n = 5$ on). Revision (v3): the $m = 10$ row certified— $p(1,2,3,4,10) = 7/22$ exactly, by an audit-driven second attempt pairing a sound search screen with an exact Lipschitz box certificate. With the program's bounded positive theorem and a certified deepest-hole archive.

Reformulation Program (companion to the pair-sum monograph)
A negative-results paper: no solution, no new cases of the conjecture; the surviving equivalent forms are as hard as they ever were.